



Document Effective Date
20/09/2023

RISK VERIFICATION GUIDELINE

DOC NO: IS-HSE-GDL-312

Rev: 01

Quality Reviewed By: OIB/4

Digitally signed by
Moklis Miah
Date: 2023.09.20
09:57:10 +03'00'

CUSTODIAN DEPARTMENT	DOCUMENT AUTHOR	DOCUMENT OWNER
HEALTH, SAFETY & ENVIRONMENT (IDD EL-SHARGI)	Mohamed A. Elazab Mahmoud Digitally signed by Mohamed A. Elazab Mahmoud DN: cn=Mohamed A. Elazab Mahmoud gn=Mohamed A. Elazab Mahmoud c=US United States e=US United States email=mahmoud@qatarenergy.qa Reason: I am the author of this document Location Date: 2023-09-20 09:24+03:00	Tigor Maruli Digitally signed by Tigor Maruli Date: 2023.09.20 09:49:59 +03'00'
	OIH/41	A/OIH/4

FINAL APPROVAL
 IBRAHIM AL-AHMAD
OIH

	RISK VERIFICATION GUIDELINE DOC NO: IS-HSE-GDL-312	REV. 01
---	---	----------------

TABLE OF CONTENTS

Page No.

1.0 PURPOSE 3

2.0 SCOPE AND APPLICABILITY..... 3

3.0 GUIDELINE 3

 3.1 GENERAL 3

 3.2 CAUSES OF IMPROPER ASSIGNMENT OF RISK LEVEL..... 4

 3.3 RISK VERIFICATION PROCESS..... 6

4.0 REFERENCES TO OTHER DOCUMENTS 10

5.0 DEFINITIONS 11

APPENDIX..... 12

 APPENDIX A - RISK VERIFICATION FORM..... 12

 APPENDIX B - REVISION HISTORY LOG 13

 APPENDIX C - LIST OF ABBREVIATIONS 13

	RISK VERIFICATION GUIDELINE DOC NO: IS-HSE-GDL-312 REV. 01
---	---

1.0 PURPOSE

This Guideline is intended to provide guidance on HSE Risk Verification, required in the Risk Evaluation and Response Procedure, IS-HSE-PRC-301.

The General objectives of the Idd El Shargi Risk Management Guidelines within the Idd El Shargi RM process are to:

- Allow a clearer definition of requirements with regard to the Idd El Shargi RM program and related Policies and Procedures;
- Allow specific topics to be addressed in detail relevant to the application of the RM element, including examples where relevant, and
- Act as a basis for further information, understanding and training.

Note: These Documents should be considered as Guidelines only and not mandatory unless stated. For further clarification on content and intent contact Idd El Shargi HSE Risk Engineer

2.0 SCOPE AND APPLICABILITY

This Guideline applies to Employees, contractors, and subcontractors, whether engaged directly by QatarEnergy or by others, who will perform work or services of any kind at any Idd El Shargi facilities and/or locations.

The Guideline applies to all personnel within Idd El Shargi who are involved in HSE Risk Assessment and Verification.

3.0 GUIDELINE

3.1 GENERAL

The HSE Risk Evaluation and Response Procedure (IS-HSE-PRC-301), requires that when a residual C Level Risk without Potential Elevated Consequence(s) PEC is identified in a Process Hazard Review (PHR), verification of the Risk Level shall be carried out by the PHR team during the review itself, and duly noted, to ensure that all potential Mitigation measures have been considered and that Consequence and Probability levels have been correctly assigned.

If a C-w/PEC, D or E Level HSE Risk is identified, an appropriately qualified Risk Evaluator from HSE Risk Engineering shall determine if:

- The Risk Level and any PEC codes have been properly developed.
- More analysis is needed to establish mitigation options.

This is required to ensure that proper Risk Evaluation of Event Scenarios is taking place and it also serves as a quality check of the evaluation.

Verifications carried out shall be documented, using the Risk Verification Form and filed in the Technical HSE Engineering Directory.

	RISK VERIFICATION GUIDELINE DOC NO: IS-HSE-GDL-312 REV. 01
---	---

3.2 CAUSES OF IMPROPER ASSIGNMENT OF RISK LEVEL

In the course of conducting Process Hazard Reviews (PHRs), and other process related activities for which potential hazards are identified, event scenarios are postulated and developed to describe how a hazard may arise and culminate in various adverse consequences. A team with diverse areas of experience is required to evaluate the risks of the event scenario and residual risk of the event scenario based on recommended mitigations. This process involves making estimates of consequences for each applicable consequence category and a corresponding probability based on the criteria provided in the Idd El Shargi Risk Matrix, IS-HSE-PRC-301, Appendix 1.

The Risk Evaluation Team is required to have sufficient experience and expertise to arrive at a reasonable estimation of the consequence levels for each Event Scenario. The Risk Evaluation Team with sufficient experience and expertise will also be able to reasonably estimate the probability associated with the consequence using the CME methodology described in IS-HSE-GDL-309. Table 1 below gives common reasons why the teams may arrive at inaccurate or indefensible Risk Levels. The Risk Verification is not carried out to determine the adequacy or effectiveness of the Risk Evaluation Team, but rather personnel carrying out the verifications should be aware of these factors that might influence the assigned Risk Level and Mitigation actions.

TABLE 1. RISK EVALUATION DEFICIENCIES IN QUALITATIVE RISK EVALUATIONS

Risk Evaluation Deficiencies	Examples
Team doesn't properly understand risk evaluation methodology	Event scenarios are not fully described. Team evaluates the probability of "loss of containment" rather than the final consequences. CMEs inappropriately credited, or not credited, for event scenarios. "Double Jeopardy" ¹ applied to CMEs.
Team does not have adequate information	Drawings contain numerous errors. MSDS, PFDs, Cause and Effects diagrams not available. Equipment repair / replacement costs and loss of revenue cost not well known.
Team constitution is missing a key role	No operator or technician on the Risk Evaluation Team.
Team is lacking expertise in process or system	New process or system that is not familiar to the team members.

	RISK VERIFICATION GUIDELINE DOC NO: IS-HSE-GDL-312 REV. 01
---	---

Risk Evaluation Deficiencies	Examples
Team Member is lacking expertise in relevant subject matter	An experienced process engineer is not knowledgeable of the specific technology or equipment that is under analysis.
Team member is lacking experience in general	Person working as a Safety Rep. has worked less than 1 year. Operator only recently moved into area that is under analysis.
Team member is lacking experience in specific relevant subject matter	Environmental Engineer specializing in oily waste is called upon for air emissions issues.
Team unduly influenced ²	Team Facilitator who is biased towards certain beliefs or directly participates rather than facilitating. Team member with an 'agenda' to do, or not do, some action.
Team not balanced ²	Team member who is outspoken controls, and is not controlled by Facilitator. Team members intimidated or not encouraged to participate.

1. Two or more unrelated events or conditions that would have to occur concurrently or be present in order to initiate the sequence of events that lead to adverse consequences. The combined frequency of occurrence is so low as to consider the event scenario "not credible". The unrelated events or conditions claimed for double jeopardy are not CMEs that protect against escalation of the Event Scenario.
2. These are subjective deficiencies which the verifier may or may not be informed by the team members. If the verifier is made aware of such issues, then particular attention should be paid to scenarios to ensure the assigned risk level is defensible.

3.3 RISK VERIFICATION PROCESS

The Risk Verification process is illustrated in the following diagram below in Figure 1:



Figure 1. Risk Verification Process

	RISK VERIFICATION GUIDELINE DOC NO: IS-HSE-GDL-312 REV. 01
---	---

3.3.1 VERIFICATION REQUIREMENT

When a C-w/PEC, D or E Level Risk is identified, a review by a competent person (as defined in 4.3.2 below), shall be carried out, to verify that the Risk Level and the PEC codes have been properly developed and if more analysis is required to establish further Mitigation options. The Verification shall be carried out as follows:

- Review of Consequence and Probability data, including effectiveness and applicability of intrinsic and/or planned 'Additional Controls' (Critical Mitigation Elements (CMEs), Risk Mitigation Measures (RMMs) and Key Mitigation Elements (KMEs), to verify the team-assigned Risk Levels
- Review the Event Scenario to determine if any further 'Additional Controls' are required and can be effectively applied.
- Revert to Proponent / Change Owner with findings and update documentation accordingly.
- The verification process may apply a level of detailed approach higher than that of the initial team analysis, e.g. Layer of Protection Analysis (LOPA), to confirm a CME/KME based qualitative assessment; unless the verification finds clear, errors in the initial approach / assumptions.

When a residual C Level Risk with no PEC code (PEC NA), is identified in a PHR, verification of the C Level Risk shall be carried out by the PHR team during the review itself, and duly noted, to ensure that all potential Mitigation measures have been considered and that Consequence and Probability levels have been correctly assigned.

Verifications carried out for C-w/PEC, D and E Level HSE Risk Levels shall be documented, using the Risk Verification Form in Appendix 1, and electronically filed in the HSE Directory.

The Idd El Shargi HSE Risk Management Coordinator or designated qualified risk evaluator shall provide verification of the Risk Level.

3.3.2 RISK VERIFIER – COMPETENCY

The appointed Risk Verifier shall be a member of the Idd El Shargi HSE Risk Engineering Team and shall possess the following minimum skill-set:

- Trained in the Idd El Shargi Risk Management program
- Trained in the CME methodology for assignment of probability to Event scenarios
- Trained in the relevant PHR methodologies
- Knowledgeable with Idd El Shargi Procedures and Guidelines especially the RM Procedures and all associated Guidelines.
- Have earned a degree in Engineering
- Have worked, or been involved, with hydrocarbon production / processing for a minimum of 3 years.

3.3.3 DEFINE SCOPE OF RISK VERIFICATION

The Risk Evaluator must ensure that all HSE Risks evaluated as C-w/PEC, D or E Level risks are verified.

3.3.4 REVIEW EVENT SCENARIO DESCRIPTION

	RISK VERIFICATION GUIDELINE DOC NO: IS-HSE-GDL-312 REV. 01
---	---

Since “C-w/PEC and higher Risk Levels are captured in the Idd El Shargi Risk Inventory, the Event Scenarios should be clearly defined and specific enough such that a person who is not directly involved with the issue would have an understanding of it.

Typically, Event Scenarios should be phrased by describing:

- a. The specific initiating event (e.g. internal corrosion, control loop failure, human error etc.)
- b. The progression of escalating events leading to loss of containment (or a description of the hazard), and the further progression that resulted in the significant adverse consequences which drives the Risk Level.

If the Event Scenarios are identified in a What-If study, the initiating events may need to be clarified, and re-phrased, from the What-If questions. If the Event Scenarios are identified in a HAZOP study, the “Cause” column will normally contain the initiating event, and the “Consequences” column should contain the escalating events and final consequences. The Risk Evaluator should ensure that the Event Scenario descriptions are clearly stated to be understandable and unambiguous.

3.3.5 REVIEW EVENT SCENARIO CONSEQUENCE CATEGORY & CONSEQUENCE LEVEL

The Risk Evaluator should examine each of the Event Scenarios in the scope of the Risk Verification to ensure that the Risk Assessment Team evaluated each of the Consequence categories. With the exclusion of security and transportation Event Scenarios, there are six Consequence categories to be verified:

- Human Injury at a Facility
- Facility Property Damage and Loss of Revenue
- Human Injury outside the Facility
- Damage to Property not owned / operated by Idd El Shargi
- Environmental Impairment: Remediation
- Environmental Impairment: Sea life.

There should be records that each of these Consequence categories had been evaluated, and it should be noted if any category was considered not to apply to the Event Scenario. If there is no evidence that the Risk Assessment Team evaluated relevant consequence categories, or if the assigned consequence level is not considered to be reasonable, the options are:

- Reconvene a Risk Assessment meeting with the full team, or an ad-hoc team, inclusive of the Risk Evaluator, to re-assess the risk.
- Employ hazard and consequence modelling software / refine potential impacts, particularly for loss of containment Event Scenarios. Typical types of hazard and consequence modelling are:
 - Liquid release rate (potential PD/LOR and Environmental impacts)
 - Flammable and Toxic Gas dispersion modelling (piping and equipment leaks and ruptures, pressure relief device releases, pool vaporization).
 - H2S and CO2 Radius of Exposure modelling

	RISK VERIFICATION GUIDELINE DOC NO: IS-HSE-GDL-312 REV. 01
---	---

- Toxic Gas dosage modelling
- Thermal Radiation modelling (pool fire, jet fire, vapor cloud fire)
- Explosion modelling (vapor cloud explosion, vessel burst, BLEVE).

3.3.6 REVIEW EVENT SCENARIO PROBABILITY DATA

For process risks, the Probability Levels are determined using the Critical Mitigation Element (CME) approach. For non-process risks, the Probability Levels will normally be evaluated by Risk Mitigation Measures (RMM). RMMs have been defined for Drilling and Well servicing risks (a specific designation of Critical Mitigation System, CMS, has been assigned to these RMMS, see IS-HSE-PRC-301, Appendix 2).

The Risk Evaluator should review each Event Scenario, including existing and planned CMEs / RMMs when applicable, with consideration to:

- Are the ultimate consequences developed without taking credit for any existing mitigations?
- Do the claimed CMEs / RMMs appear to be reasonable and effective in significantly reducing the probability of the consequences or preventing the event?
- Are the CMEs / RMMs independent of each other?
 - Verify independency of sensing elements, logic solvers and final elements (e.g. valves, switches etc.) Note: Two CMEs may be credited for independent control loops that are in a DCS, provided the control system itself is sufficiently reliable).
 - Human actions credited only once.
- Are more CMEs / RMMs claimed than is defensible? (e.g. scenario claims Fireproofing, Fire Protection and Alert and Escape System / Factors for the same Event Scenario).
- Are there other CMEs / RMMs that should have been credited?

Further guidance on CMEs is given in IS-HSE-GDL-309, Critical Mitigation Element (CME) Methodology Guideline.

3.3.7 REVIEW ASSIGNED RISK LEVEL

Using the Idd El Shargi Risk Matrix, IS-HSE-PRC-301, Appendix 1, the Risk Evaluator should verify that the assigned Risk Level for each Event Scenario is correct for the Consequence and Probability Levels developed. The residual Risk Level, based on the proposed CMEs / RMMs should be verified.

3.3.8 REVIEW PROPOSED MITIGATION ACTION PLANS

The Risk Evaluator should review the Mitigation plans and verify that the planned Mitigations are appropriately designated as CMEs, RMMs or KMEs.

3.3.9 DOCUMENT RISK VERIFICATION

All Verifications carried out should be documented by the Risk Evaluator; Appendix 1 shows the Verification template to be utilized. Any changes to the credited CMEs, KMEs, Risk Levels or recommended actions made by the Risk Evaluator should be clearly described and the basis for making the changes documented.

3.3.10 MANAGEMENT APPROVAL

	RISK VERIFICATION GUIDELINE DOC NO: IS-HSE-GDL-312 REV. 01
---	---

The completed verification form shall be signed off by the Risk Evaluator and approved by the Idd El Shargi, HSE manager, Field Operations Manager and the Asset Manager.

All verified C-w/PEC, D and E HSE Risk Levels shall be submitted to Idd El Shargi HSE Risk Management for Risk Validation, prior to submission to the Management.

Verified C-w/PEC, D and E Level Risks will be documented in the Idd El Shargi HSE Risk Registers, as defined in IS-HSE-PRC-301.

The verified risks, documented in the Idd El Shargi risk registers during the year, shall be tabulated and incorporated in the Annual Risk Inventory.

4.0 REFERENCES TO OTHER DOCUMENTS

TABLE 2: REFERENCES TO OTHER DOCUMENTS

Document Title	Document Number	Internal/External Document (if any)	Document Referencing
Risk Evaluation and Response Procedure	IS-HSE-PRC-301	Internal	Upwards
Critical Mitigation Element (CME) Methodology Guideline	IS-HSE-GDL-309	Internal	Sideways

	RISK VERIFICATION GUIDELINE DOC NO: IS-HSE-GDL-312 REV. 01
---	---

5.0 DEFINITIONS

TABLE 3: KEY TERMS DEFINED

Term	Definition
CME	Critical Mitigation Element, An independent and reliable device or system that is capable of: 1) preventing initiation of an Event Scenario, 2) preventing loss of Hazardous Material containment, or 3) substantially mitigating one of an Event Scenario's Consequences. Failure of one CME shall not prevent another CME from performing its function. CME shall be maintained to the reliability level assumed in the risk analysis. A CME reduces an Event Scenario's Probability by one level on the Risk Matrix.
HAZOP	Hazard and Operability Study A formal team-based, structured, hazard identification and risk assessment method that applies a detailed deviation, guideword, parameter, cause, consequence, and controls approach. Its main objective being to validate a robust design in terms of Hazards And Operability- it is not intended as a design review.
KME	Key Mitigation Element HSE Risk mitigation elements (equipment, design features, specifications, inspection and maintenance routines, and operating parameters) that are necessary to achieve and maintain the reliability level of Critical Mitigation Elements (CMEs), and Risk Mitigation Measures (RMM), or risk mitigations that are not otherwise claimed as CMEs / RMMs in a risk analysis.
PEC	Potential Elevated Consequences Consequences of an Event Scenario that exceed 14 Permanent Incapacitating Injuries or \$1 billion of combined PD/LOR and thereby substantially exceed the threshold for Consequence Level 5.
PII	Permanent Incapacitating Injury A fatality or physical injury or illness that permanently renders a person incapable of performing the essential duties of any of the occupations for which a person is qualified by education, training and experience
PHR	Process Hazard Review A systematic approach for identifying and assessing Process Hazards and risks and their possible effects.
RMM	Risk Mitigation Measures A group of mitigation measures that are given credit for reducing the Probability in the Risk Matrix by one level (e.g., Probability Level 5 to Probability Level 4) in risk evaluations where the Critical Mitigation Element (CME) methodology cannot be applied.

DOC NO: IS-HSE-GDL-312

REV. 01

APPENDIX A - RISK VERIFICATION FORM

Rev.	Date	Description/Source	Verified By:	☐ Approved ☐ Rejected	☐ Approved ☐ Rejected	☐ Approved ☐ Rejected			
1. LOCATION				2. IDENTIFICATION					
1.1 <u>Asset:</u>				2.1 <u>Change/Study ID:</u>		2.2 <u>Risk Status:</u>			
1.2 <u>Facility/System:</u>				2.3 <u>Title:</u>					
3. HAZARD SCENARIO									
3.1 <u>Operating Mode:</u>		3.2 <u>Hazard:</u>		3.5 <u>Consequence Categories: (Check all that apply)</u>					
3.3 <u>Initiating/Escalating Events:</u>				☒ Personnel					
				☒ Property and Revenue					
3.4 <u>Consequences:</u>				☒ Environmental					
				☒ Impacts Outside a Facility					
4. RISK VERIFICATION									
4.1 <u>Strategy</u>		4.2 <u>Method of Approach:</u>							
4.3 <u>Discussion: (Note: Add supportive data to attachment sheet as required)</u>									
4.4 <u>Conclusion / Recommendations:</u>									
5. RISK RANK AFTER VERIFICATION									
5.1 Initial Consequence: X				5.3 Initial Risk Rank:		5.4 Residual Consequence: X			5.6 Residual Risk Rank:
5.2 Initial Probability: X									
5.10 Comments:									

Attach the Revision B report here, together with any other supporting documentation.

	RISK VERIFICATION GUIDELINE DOC NO: IS-HSE-GDL-312 REV. 01
---	---

APPENDIX B - REVISION HISTORY LOG

TABLE 4: REVISION HISTORY LOG

Revision Number: 01

Date: 18/09/2023

Item Revised:	Revision Description	Page No.
All	Document simplified as part of QMR initiative.	All
Remarks:		

APPENDIX C - LIST OF ABBREVIATIONS

TABLE 5: LIST OF ABBREVIATIONS

Abbreviation	Description
CME	Critical Mitigation Element
HAZOP	Hazard and Operability Study
IPL	Independent Protection Layers
KME	Key Mitigation Element
LOPA	Layers of Protection Analysis
MSDS	Material Safety Data Sheets
PEC	Potential Elevated Consequences
PFD	Process Flow Diagram
PHR	Process Hazard Review
PII	Permanent Incapacitating Injury
RMM	Risk Mitigation Measures